### Illustration 10.1

Let us make an attempt to find the slope of the tangent line to the graph of $f(x) = x^2$ at (1, 1).

As a start, let us take $\Delta x = 0.1$ and find the slope of the secant line through $(1,1)$ and $(1.1,(1.1)^2)$.

(i) $f(1.1) = (1.1)^2 = 1.21$

(ii) $\Delta y = f(1.1) - f(1)$

$= 1.21 - 1 = 0.21$

(iii) $\dfrac{\Delta y}{\Delta x} = \dfrac{0.21}{0.1} = 2.1$

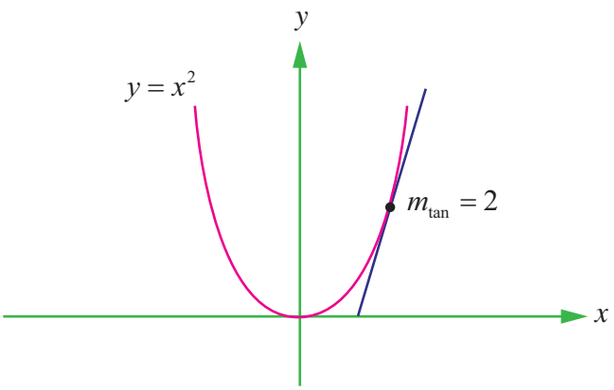



*Fig. 10.14*

Tabulate the successive values to the right and left of 1 as follows :

| $\Delta x$ | $1 + \Delta x$ | $f(1)$ | $f(1 + \Delta x)$ | $\Delta y$ | $\Delta y / \Delta x$ |
|---|---|---|---|---|---|
| 0.1 | 1.1 | 1 | 1.21 | 0.21 | 2.1 |
| 0.01 | 1.01 | 1 | 1.0201 | 1.0201 | 2.01 |
| 0.001 | 1.001 | 1 | 1.002001 | 0.002001 | 2.001 |
| – 0.1 | 0.9 | 1 | 0.81 | – 0.19 | 1.9 |
| – 0.01 | 0.99 | 1 | 0.9801 | – 0.0199 | 1.99 |
| – 0.001 | 0.999 | 1 | 0.998001 | – 0.001999 | 1.999 |

Clearly, $\lim\limits_{\Delta x \to 0^-} \dfrac{\Delta y}{\Delta x} = 2$ ; $\lim\limits_{\Delta x \to 0^+} \dfrac{\Delta y}{\Delta x} = 2$.

This shows that $\lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x} = 2$ .

Thus the slope of the tangent line to the graph of $y = x^2$ at (1, 1) is $m_{tan} = 2$.

On the basis of the Fig.10.6 to 10.13, Illustration 10.1, and our intuition, we are prompted to say that if a graph of a function $y = f(x)$ has a tangent line $L$ at a point $P$, then $L$ must be the line that is the **limit** of the secants $PQ$ through $P$ and $Q$ as $Q \to P$ ($\Delta x \to 0$). Moreover, the slope $m_{tan}$ of $L$ should be the **limiting** value of the values $m_{sec}$ as $\Delta x \to 0$. This is summarised as follows:

**Definition 10.1 (Tangent line with slope *m*)**

Let $f$ be defined on an open interval containing $x_0$, and if the limit

$\lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = m_{tan}$ exists, then the line passing through $(x_0, f(x_0))$ with slope $m$ is the **tangent line** to the graph of $f$ at the point $(x_0, f(x_0))$.

The slope of the tangent line at $(x_0, f(x_0))$ is also called the **slope of the curve** at that point.

The definition implies that if a graph admits tangent line at a point $(x_0, f(x_0))$ then it is unique since a point and a slope determine a single line.



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